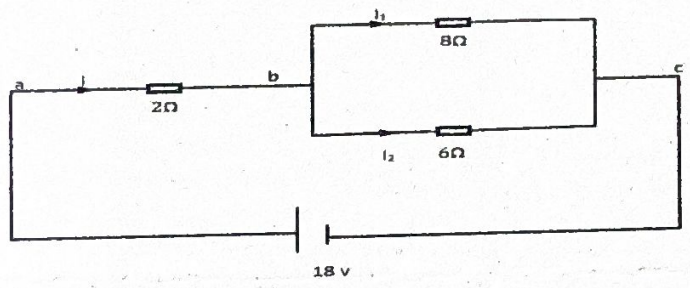
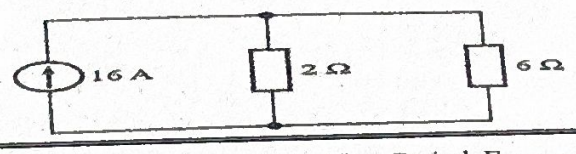
Internal Assessment - I (IA-1)

Department: EEE
Course Name: Basic Electrical Engineering
Batch: 2025-26
Semester: II

Date: 19/05/2026
Course Code: 25BEPHY205
Max. Marks: 50
Duration: 90 Mins

Note: Answer all the questions. Q1 & Q2 carries 10 Marks each, Q3 - Q6 carry 20 Marks each

Q No	Question	Marks	RBT Level	CO
1.	a. Explain Faradays laws of electromagnetic induction	5	L2	CO1
	b. Derive the expression for Average Value of an alternating current and voltage.	5	L2	CO2
	OR			
2.	a. State and explain Ohm's law with its limitations	5	L2	CO1
	b. Derive the expression for Form Factor and Peak Factor of Sinusoidal Waveforms	5	L2	CO2
3.	a. Explain the structure of Power system along with single line diagram	8	L2	CO1
	b. Illustrate KCL and KVL with an Example	6	L2	CO2
	c. Determine I, I_1, I_2 , V_{ab} and V_{bc} in the network of fig. 	6	L3	CO3
	OR			
4.	a. Discuss the current division rule of DC Circuits	6	L2	CO1
	b. Describe voltage and current source transformations with neat sketch	6	L2	CO1
	c. Find the currents through the two resistors in the circuit of fig. Then transform the current source and 2- Ω resistor to an equivalent voltage source and again find the currents through the two resistors. Compare the results. 	8	L3	CO3
5.	a. Define the following terms with diagram: Cycle, Time Period, Frequency, Amplitude, and Instantaneous value of a Sinusoidal signal with neat sketch.	6	L2	CO1
	b. Describe voltage, current and power in pure resistance AC circuit with neat sketch	8	L2	CO1
	c. Calculate the expression for the sum of the currents : $i_1 = 10\sqrt{2}\sin \omega t$ A and $i_2 = 20\sqrt{2}\sin(\omega t + 60°)$ A . Also, determine the rms value of the sum of these two currents	6	L3	CO3
	OR			
6.	a. Derive the expression for RMS value of of an alternating current and voltage.	6	L2	CO2
	b. Discuss voltage, current and power in pure inductance AC circuit with neat sketch	8	L2	CO1
	c. Following three sinusoidal currents flow into a junction: $i_1 = 5\sin \omega t$, $i_2 = 5\sin(\omega t + 30°)$, $i_3 = 5\sin(\omega t - 120°)$. Calculate the rms value of the resultant current (expressed in polar form) that leaves the junction.	6	L3	CO3